

```
Processando config 21, seed 0, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
```

```
Correction factor of the ChiSquare:
```

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 3.46
p-value of test : 0.18
```

```
-----
CPU time used : 00:00:00.03
```

```
Generator state:
```

```
16777216 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 5.43
p-value of test : 0.71

CPU time used : 00:00:00.02

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 138.13
p-value of test : 0.26

CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32655.21
p-value of test : 0.67

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 179.55
p-value of test : 0.43

CPU time used : 00:00:00.03

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 295.30
p-value of test : 0.64

CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.10
p-value of test : 0.46

CPU time used : 00:00:00.02

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.97
p-value of test	:	0.47

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	20.40
p-value of test	:	0.93

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	36.59
p-value of test	:	0.26

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	20.15
p-value of test	:	0.74

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	10.39
p-value of test	:	0.79

CPU time used : 00:00:00.08

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	44.36
p-value of test	:	0.96

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	65.27
p-value of test	:	0.36

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	176.61
p-value of test	:	0.17

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	61.18
p-value of test	:	0.33

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	32.58
p-value of test	:	0.44

CPU time used : 00:00:00.06

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.31

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic            : 0.93
p-value of test                   : 0.63
```

```
-----
CPU time used                     : 00:00:00.03
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	23.09
p-value of test	:	3.2e-3

CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	97.62
p-value of test	:	0.98

CPU time used : 00:00:00.00

Generator state:

201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 0.00097653271,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 32768
Value of the statistic            : 32378.41
p-value of test                   : 0.94
```

```
-----
CPU time used                     : 00:00:00.03
```

Generator state:

218103808 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

sstring_HammingIndep test:

```
-----
N = 1,  n = 524288,  r = 0,  s = 32,  L = 16,  d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom      : 177
Chi-square statistic             : 205.97
p-value of test                   : 0.07
```

```
-----
CPU time used                     : 00:00:00.02
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	334.26
p-value of test	:	0.12

CPU time used	:	00:00:00.02
---------------	---	-------------

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-1.96
p-value of test	:	0.97

CPU time used	:	00:00:00.02
---------------	---	-------------

Generator state:

268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	32.28
p-value of test	:	0.45

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	36.85
p-value of test	:	0.22

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	39.09
p-value of test	:	0.18

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	24.29
p-value of test	:	0.50

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	19.85
p-value of test	:	0.18

CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	72.46
p-value of test	:	0.17

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	55.31
p-value of test	:	0.71

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	179.65
p-value of test	:	0.14

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	60.92
p-value of test	:	0.34

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	31.61
p-value of test	:	0.49

CPU time used : 00:00:00.05

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

Version:	TestU01 1.2.3
Generator:	ufire_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.28

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic            : 4.55
p-value of test                   : 0.10
```

```
-----
CPU time used                     : 00:00:00.02
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 11.57
p-value of test : 0.17

CPU time used : 00:00:00.03

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 100.67
p-value of test : 0.96

CPU time used : 00:00:00.02

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32778.17
p-value of test : 0.48

CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 192.61
p-value of test : 0.20

CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	323.79
p-value of test	:	0.22

CPU time used : 00:00:00.00

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	0.56
p-value of test	:	0.29

CPU time used : 00:00:00.02

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	36.90
p-value of test	:	0.25

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	24.75
p-value of test	:	0.78

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	25.93
p-value of test	:	0.77

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	26.19
p-value of test	:	0.40

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	10.30
p-value of test	:	0.80

CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	80.08
p-value of test	:	0.06

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	73.24
p-value of test	:	0.16

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	156.77
p-value of test	:	0.56

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	64.38
p-value of test	:	0.23

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	24.70
p-value of test	:	0.82

CPU time used : 00:00:00.05

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216

Number of statistics: 17
Total CPU time: 00:00:00.27

All tests were passed

XXX
Starting Alhabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 7.79
p-value of test : 0.02

CPU time used : 00:00:00.02

Generator state:
469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic       : 6.01
p-value of test              : 0.65
```

CPU time used : 00:00:00.03

Generator state:

486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic       : 132.65
p-value of test              : 0.37
```

CPU time used : 00:00:00.00

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32741.75
p-value of test : 0.54

CPU time used : 00:00:00.03

Generator state:

520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 173.38
p-value of test : 0.56

CPU time used : 00:00:00.02

Generator state:
536870656 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	: 305
Chi-square statistic	: 296.20
p-value of test	: 0.63

CPU time used : 00:00:00.00

Generator state:
553647872 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 7.25e-3
p-value of test : 0.50

CPU time used : 00:00:00.00

Generator state:
570425088 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

```
-----
N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64
```

Test on the values of the Statistic H

```
Number of degrees of freedom      : 32
ChiSquare statistic                : 26.90
p-value of test                   : 0.72
```

Test on the values of the Statistic M

```
Number of degrees of freedom      : 31
ChiSquare statistic                : 35.76
p-value of test                   : 0.25
```

Test on the values of the Statistic J

```
Number of degrees of freedom      : 32
ChiSquare statistic                : 26.84
p-value of test                   : 0.73
```

Test on the values of the Statistic R

```
Number of degrees of freedom      : 25
ChiSquare statistic                : 25.52
p-value of test                   : 0.43
```

Test on the values of the Statistic C

```
Number of degrees of freedom      : 15
ChiSquare statistic                : 10.48
p-value of test                   : 0.79
```

CPU time used : 00:00:00.08

Generator state:
587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	54.42
p-value of test	:	0.74

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	62.43
p-value of test	:	0.46

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	202.49
p-value of test	:	0.01

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	58.62
p-value of test	:	0.42

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	21.14
p-value of test	:	0.93

CPU time used : 00:00:00.05

Generator state:
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 14.93
p-value of test : 5.7e-4 *****

CPU time used : 00:00:00.02

Generator state:
620756480 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 10.15
p-value of test : 0.25

CPU time used : 00:00:00.00

Generator state:
637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 138.71
p-value of test : 0.24

CPU time used : 00:00:00.02

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32731.49
p-value of test : 0.56

CPU time used : 00:00:00.03

Generator state:

671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 170.09
p-value of test : 0.63

CPU time used : 00:00:00.02

Generator state:

687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305

Chi-square statistic : 317.58

p-value of test : 0.30

CPU time used : 00:00:00.00

Generator state:

704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.27

p-value of test : 0.61

CPU time used : 00:00:00.02

Generator state:

721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	37.83
p-value of test	:	0.22

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	34.97
p-value of test	:	0.28

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	33.33
p-value of test	:	0.40

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	24.53
p-value of test	:	0.49

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	17.16
p-value of test	:	0.31

CPU time used : 00:00:00.06

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	48.57
p-value of test	:	0.89

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	62.29
p-value of test	:	0.47

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	175.37
p-value of test	:	0.19

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	43.07
p-value of test	:	0.91

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	30.41
p-value of test	:	0.55

CPU time used : 00:00:00.05

Generator state:
754973952 bits have been read.

===== Summary results of Alphabit =====

```

Version:          TestU01 1.2.3
Generator:        ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:   16777216
Number of statistics: 17
Total CPU time:   00:00:00.24
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

```

Test	p-value
-----	-----
1 MultinomialBitsOver, L = 2	5.7e-4
-----	-----

All other tests were passed

```

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```

ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 32

```

smultin_MultinomialBitsOver test:

```

-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

```

```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

```

```

Number of degrees of freedom      : 2
Value of the statistic            : 7.40
p-value of test                   : 0.02

```

```

-----
CPU time used                     : 00:00:00.03

```

```

Generator state:
771751168 bits have been read.

```

```

*****

```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 8.82
p-value of test : 0.36

CPU time used : 00:00:00.02

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 119.95
p-value of test : 0.68

CPU time used : 00:00:00.00

Generator state:
805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33289.37
p-value of test : 0.02

CPU time used : 00:00:00.03

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 178.88  
p-value of test                   : 0.45  
-----
```

```
-----  
CPU time used                     : 00:00:00.00  
-----
```

Generator state:

838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 291.24  
p-value of test                   : 0.70  
-----
```

```
-----  
CPU time used                     : 00:00:00.02  
-----
```

Generator state:

855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                   : 0.73  
p-value of test                   : 0.23  
-----
```

CPU time used : 00:00:00.02

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	25.07
p-value of test	:	0.80

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	35.92
p-value of test	:	0.25

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	34.61
p-value of test	:	0.34

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.70
p-value of test	:	0.28

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	33.23
p-value of test	:	4.4e-3

CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config21_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	51.28
p-value of test	:	0.83

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	65.81
p-value of test	:	0.35

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	140.94
p-value of test	:	0.86

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	51.61
p-value of test	:	0.68

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	24.66
p-value of test	:	0.82

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed